Problem 5

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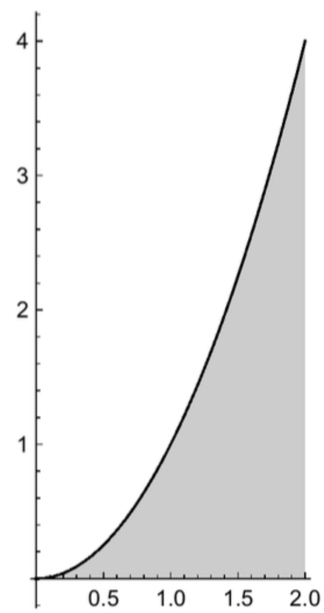
Problem 1 *The graph of the function $f(x) = x^2$ is given in the figure.*



Find the exact value of the area of the shaded region. HINT: Use a right Riemann sum with n rectangles, and then take the limit as $n \to \infty$.

Explanation. $\Delta x = \dfrac{2}{n}$

$x_k^* = x_k = k\Delta x = k\left(\dfrac{2}{n}\right) = \dfrac{2k}{n}$, for $k = 1, ..., n$

$$A \approx \sum_{k=1}^{n} f(x_k^*)\Delta x = \Delta x \cdot \sum_{k=1}^{n} f(x_k^*) = \frac{2}{n} \cdot \sum_{k=1}^{n} f\left(\frac{2k}{n}\right) = \frac{2}{n} \cdot \sum_{k=1}^{n} \left(\frac{2k}{n}\right)^2 = \frac{2}{n} \cdot \sum_{k=1}^{n} \frac{4k^2}{n^2} = \frac{8}{n^3} \cdot \sum_{k=1}^{n} k^2$$

Recall: $\displaystyle\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$

Therefore

$$A \approx \frac{8}{n^3} \cdot \sum_{k=1}^{n} k^2 = \frac{8}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{4}{3} \cdot \frac{(n+1)(2n+1)}{n^2} = \frac{4}{3} \cdot \frac{2n^2+3n+1}{n^2}.$$

Let's take the limit of Riemann sums as $n \to \infty$.

$$A = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x = \lim_{n\to\infty} \frac{4}{3} \cdot \frac{2n^2+3n+1}{n^2} = \lim_{n\to\infty} \frac{4}{3} \cdot \left(2 + \frac{3}{n} + \frac{1}{n^2}\right) = \frac{8}{3}.$$